Project Instructions

Advanced Topics in Machine Learning
SA 2025-2026

Project Overview

One mandatory group project (required to pass).

Choose between two types of projects:

1. **Reproducibility Challenge**

   Replicate results from top ML papers

2. **Research Projects**

   Produce a research-oriented scientific paper (instructor approval required)

Special rules apply to **PhD students** (see at the end)

Option 1: Reproducibility Challenge

Replicate published results from provided list of papers

You may propose different papers (from same venues, **requires TAs approval**)

Focus on **understanding** the core contribution and **critically** validating results

See instructions provided in ML Reproducibility Challenge (https://reproml.org/)

Option 2: Research Projects

Research-oriented assignment

Aim to produce a paper-like outcome with **novel contributions**

- More like a small thesis project

You have to propose your own project, **requires TAs approval**

Group Formation

Make groups of **3 members**

We prefer heterogeneous groups in terms of genders and nationality

There'll be a "looking for teammates" section for those that need to find a group

If needed, there will be 1 or 2 groups of 4 members (**do not** create 4-member groups by your own - TA will organize them)

- in this case, a **more extensive** study is expected (e.g., more experiments, additional analyses, etc)

Final remark: for **any** problem, contact the TAs

Project Guidelines

Steps for completing the project:

- **Understand** core concepts
- Perform and validate **experiments**
  - if code is available, a more extensive study is expected.
- Produce a **report**
  - **4-8 pages**, concise, avoid copy-pasting or intensive use of LLMs
- Prepare a **presentation and (optional) demo**
  - **20 minutes**, all group members must participate

The presentation will be followed by **10 minutes of Q&A**

Final Deliverables

Deadline: Submit report and code by **Monday, May 25 - 23:59**

Grading based on:

- Quality of the report
- Clarity of the presentation
- Participation in Q&A

The demo is optional (unfeasible in some case) but positively evaluated

Group effort evaluated based on team size

Experiment logging

There are several collaborative services for **experiment tracking**.

Most options offer a **free tier** for academics:

- **Weights and Biases (https://wandb.ai/)**
  - Academic tier: https://wandb.ai/site/solutions/scientific-research/
- **Comet.ml (https://www.comet.com/site/)**
  - Academic tier: https://www.comet.com/signup?plan=academic

Team registration

Choose the project you intend to do on the **iCorsi form**

Students opting for the same choice will be considered a group

A special choice is provided for students looking for teammates

When to make groups: from **March 19** to **March 31 - 23:59**

Email TAs <u>(all of us)</u> for any questions or support.

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PhD Student Rules

You have **two possibilities** to pass the course:

1. Group project (as any other student)
2. **Present** a recent machine learning **paper** of your choice
   - Ideally, related to the topics of the course and of interest for your research
   - Work on your own
   - Presentation in a different session(s) (TBA)

Either way, please **fill in your choice** on iCorsi